

Independent events



1

a $\frac{1}{6}$

b Independent

2 Dependent

3 $\frac{25}{121}$

4 $\frac{1}{4}$

5

a $\frac{1}{18}$

b $\frac{1}{36}$

6

a $\frac{1}{2}$

b $\frac{1}{2}$

c Independent

7

a $\frac{1}{2}$

b $\frac{1}{32}$

8

a $\frac{1}{2}$

b Independent

c Equally likely

9

a $\frac{64}{243}$

b $\frac{8}{243}$

c $\frac{1}{729}$

10

a Once

b $\frac{1}{6}$

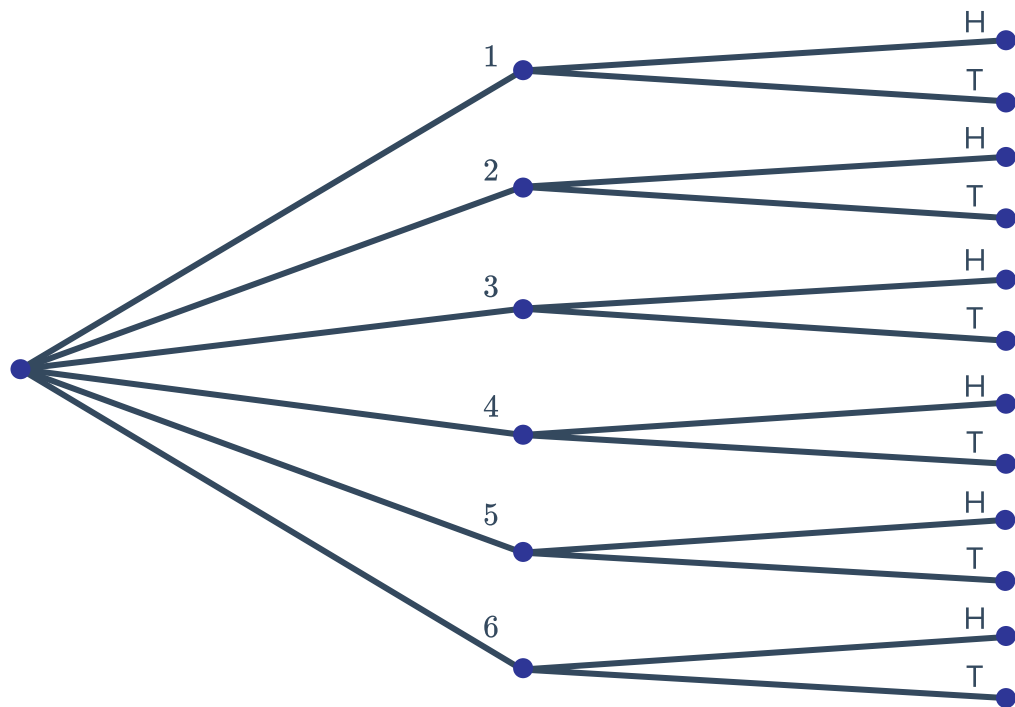
c $\frac{25}{216}$



Tree diagrams

11

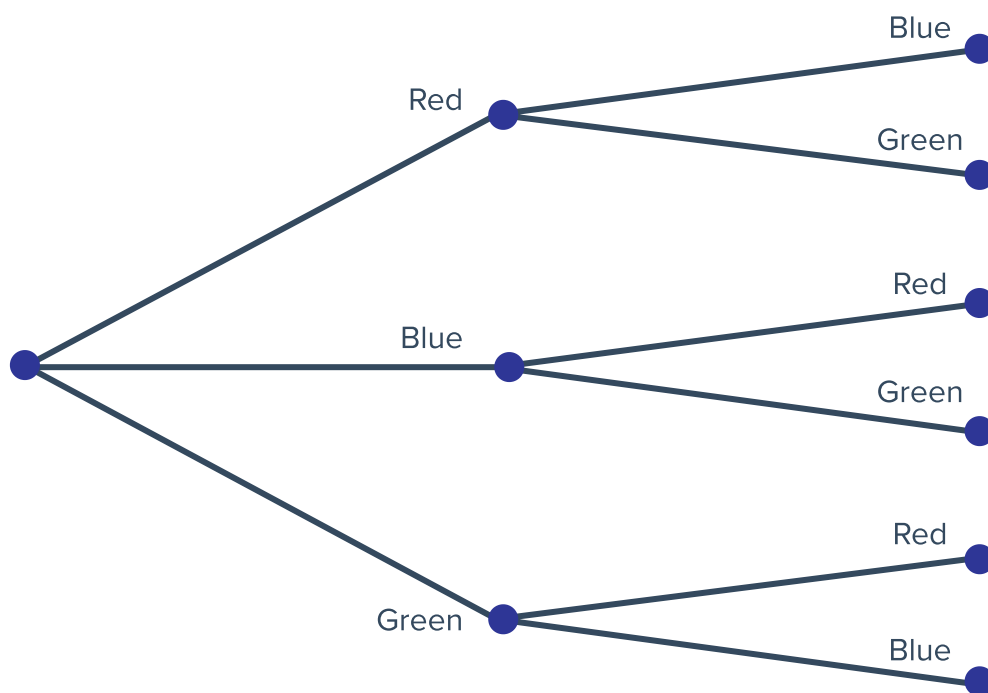
a



b $\frac{1}{4}$

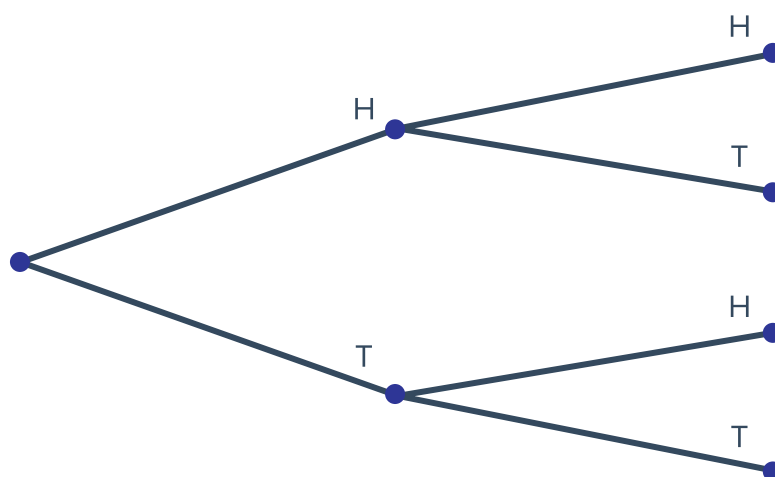
c $\frac{3}{4}$

12
a



- b $\frac{1}{3}$
 c $\frac{2}{3}$
 d $\frac{5}{9}$

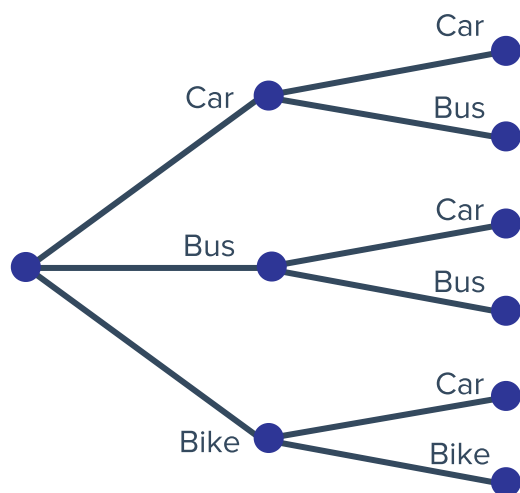
13
a



- b $\frac{1}{4}$
 c $\frac{3}{4}$

14

a



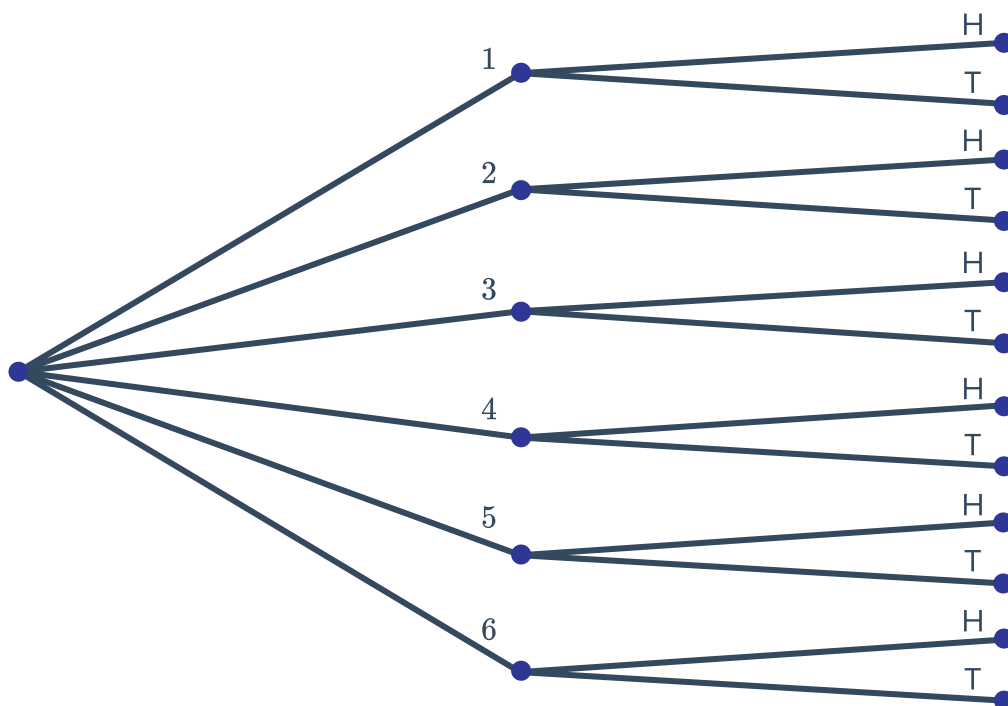
b 6

c

i $\frac{1}{2}$ ii $\frac{1}{3}$ iii $\frac{5}{6}$

15

a

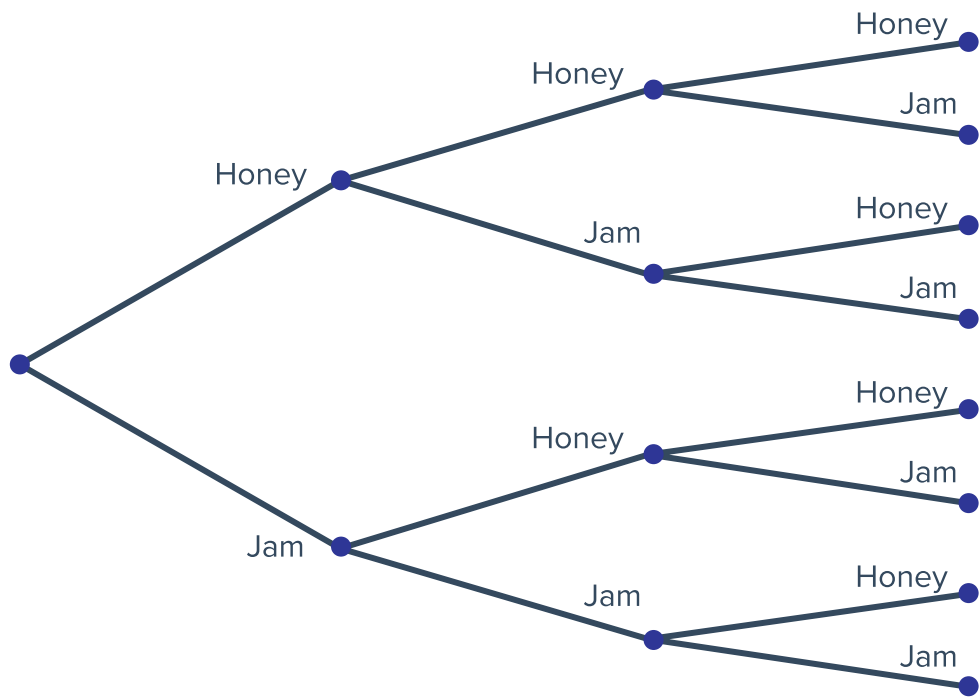


b 1H, 1T, 2H, 2T, 3H, 3T, 4H, 4T, 5H, 5T, 6H, 6T

c $\frac{1}{2}$ d $\frac{5}{12}$

16

a

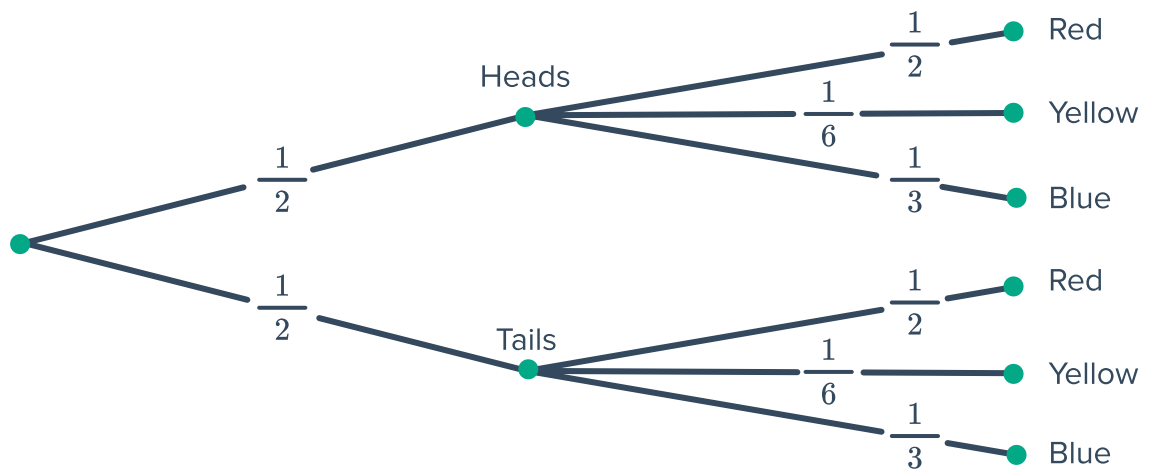


b $\frac{1}{2}$

c $\frac{1}{8}$

17

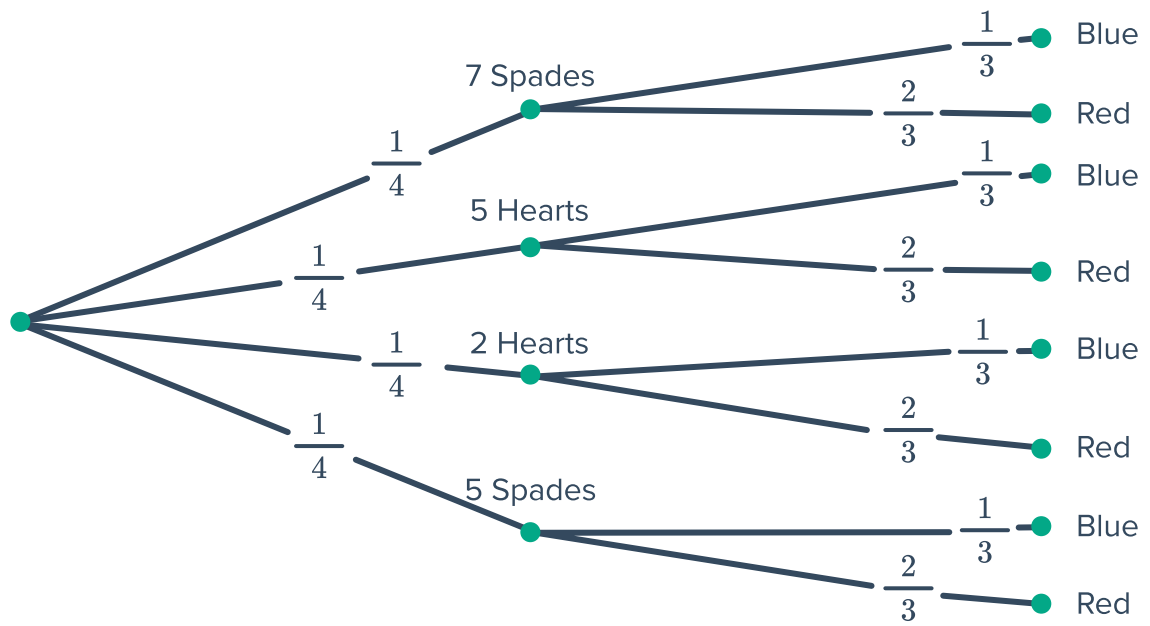
a



b $\frac{1}{12}$

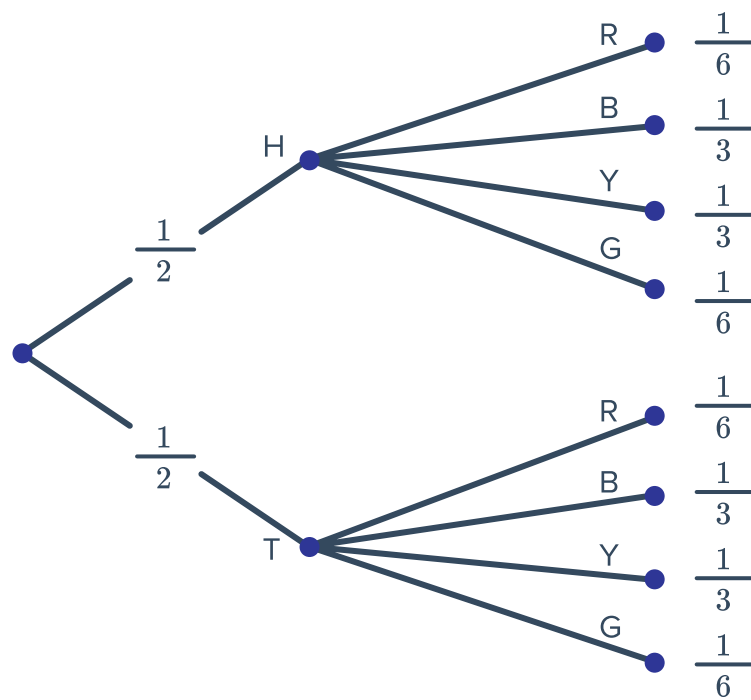
c $\frac{7}{12}$

18
a



b $\frac{1}{12}$
c $\frac{1}{2}$

19
a



b $\frac{1}{6}$
c $\frac{2}{3}$

